

Car Price Prediction Using Multiple Linear Regression

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Project: Car Price Prediction Using Multiple Linear Regression

Type: Individual Mini Project

Method: Supervised Learning

Programming Language: Python

1. Problem Statement

1.1 Introduction

The price of a used car depends on several factors such as its manufacturing year, kilometres driven, mileage, engine capacity, maximum power and number of seats. Estimating the correct selling price manually can be difficult because multiple factors affect the price at the same time. This project uses **Supervised Machine Learning** to predict the selling price of a used car based on its characteristics.

1.2 Problem Statement

The objective of this project is to develop a **Multiple Linear Regression** model that can predict the selling price of a used car using the following features:

- Manufacturing Year
- Kilometres Driven
- Mileage
- Engine Capacity
- Maximum Power
- Number of Seats

The predicted price can help understand how different vehicle characteristics are related to the selling price.

1.3 Project Objectives

The main objectives of this project are:

1. To understand the factors affecting used-car prices.
2. To clean and preprocess the car dataset.
3. To perform exploratory data analysis.
4. To study relationships between car features and selling price.
5. To apply Multiple Linear Regression.

6. To evaluate the performance of the model.
 7. To interpret the model results in simple terms.
 8. To demonstrate the application of statistical and machine learning concepts using Python.
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2. Dataset Description

2.1 Dataset

The project uses a publicly available **used-car dataset** containing information about different cars and their selling prices.

The Python program loads the dataset from a CSV file and examines its structure using Pandas.

The program determines the number of observations and variables using: `df.shape`

It also displays the column names and dataset information using `df.columns` and `df.info()`.

2.2 Variables Used

The project uses the following variables for prediction:

Variable	Description
year	Manufacturing year of the car
km_driven	Kilometres driven by the car
mileage(km/ltr/kg)	Mileage of the car
engine	Engine capacity
max_power	Maximum power of the car
seats	Number of seats
selling_price	Selling price of the car

The six input variables are used to predict the target variable **selling_price**.

2.3 Target Variable

The target variable is: `selling_price`

Selling Price (`selling_price`)

The machine learning model learns the relationship between the input features and the selling price and then predicts prices for previously unseen data.

3. Data Preprocessing

Data preprocessing is an important step because machine learning models require clean and suitable data.

3.1 Checking Dataset Information

The project first checks:

- Number of observations
- Number of variables
- Column names
- Data types
- General dataset information

The Python code uses Pandas functions such as `shape`, `columns` and `info()`.

3.2 Checking Duplicate Rows

Duplicate rows are checked using:

```
df.duplicated().sum()
```

This helps identify whether the dataset contains repeated records.

3.3 Converting Maximum Power

The `max_power` column is converted into a numeric format using:

```
pd.to_numeric(df["max_power"], errors="coerce")
```

Values that cannot be converted to numbers are changed to missing values.

3.4 Handling Missing Values

Rows containing missing values in important columns are removed.

The columns considered are:

- `selling_price`
- `year`
- `km_driven`
- `mileage(km/ltr/kg)`
- `engine`
- `max_power`
- `seats`

The project uses `dropna()` to remove incomplete records.

3.5 Feature Selection

The following six variables are selected as independent variables:

`year`

`km_driven`, `mileage(km/ltr/kg)`, `engine`, `max_power`, `seats`

The dependent variable is:

selling_price

3.6 Categorical Variables and Outliers

The final model uses numerical variables only, so categorical variables are not included in the selected feature set.

The provided preprocessing code does not apply a separate outlier-removal technique. Therefore, possible extreme values are retained in the dataset.

This is a limitation that can be considered for future improvement.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand the dataset and identify relationships between variables.

The project uses **Matplotlib** and **Seaborn** for visualization.

4.1 Distribution of Selling Prices

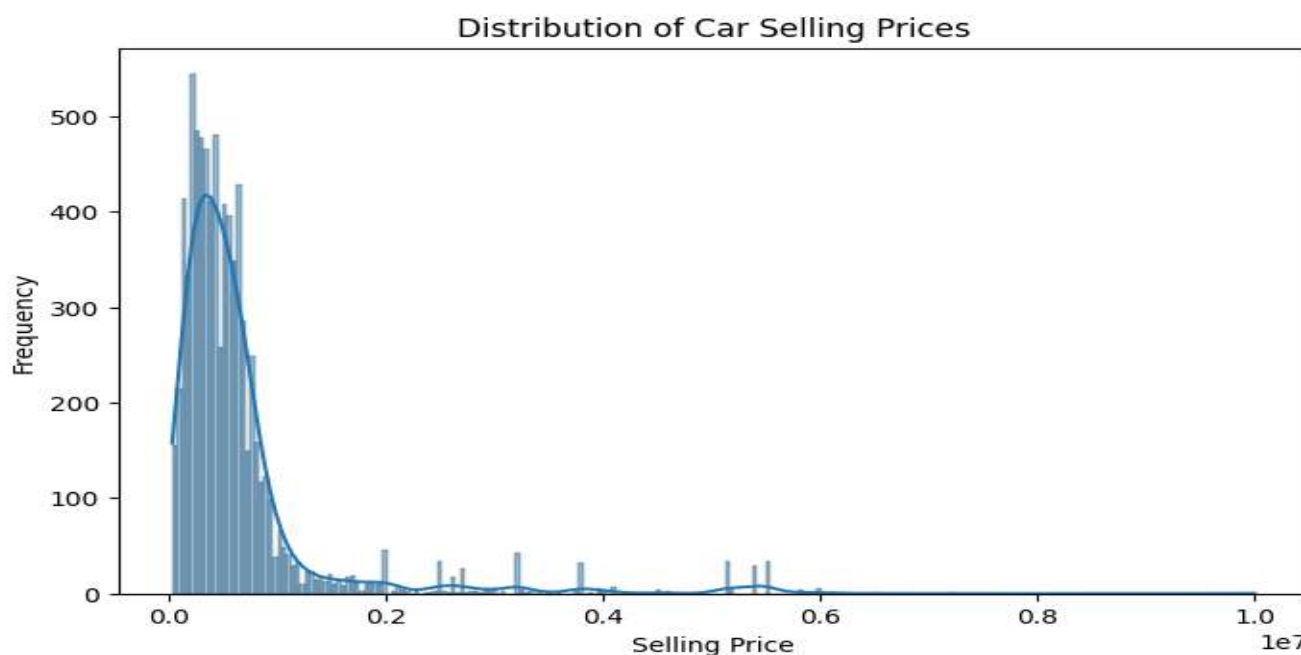
A histogram with a KDE curve is used to understand the distribution of car selling prices.

The graph helps identify:

- The general distribution of prices
- The concentration of observations
- Possible variation in selling prices

The visualization is titled:

Distribution of Car Selling Prices

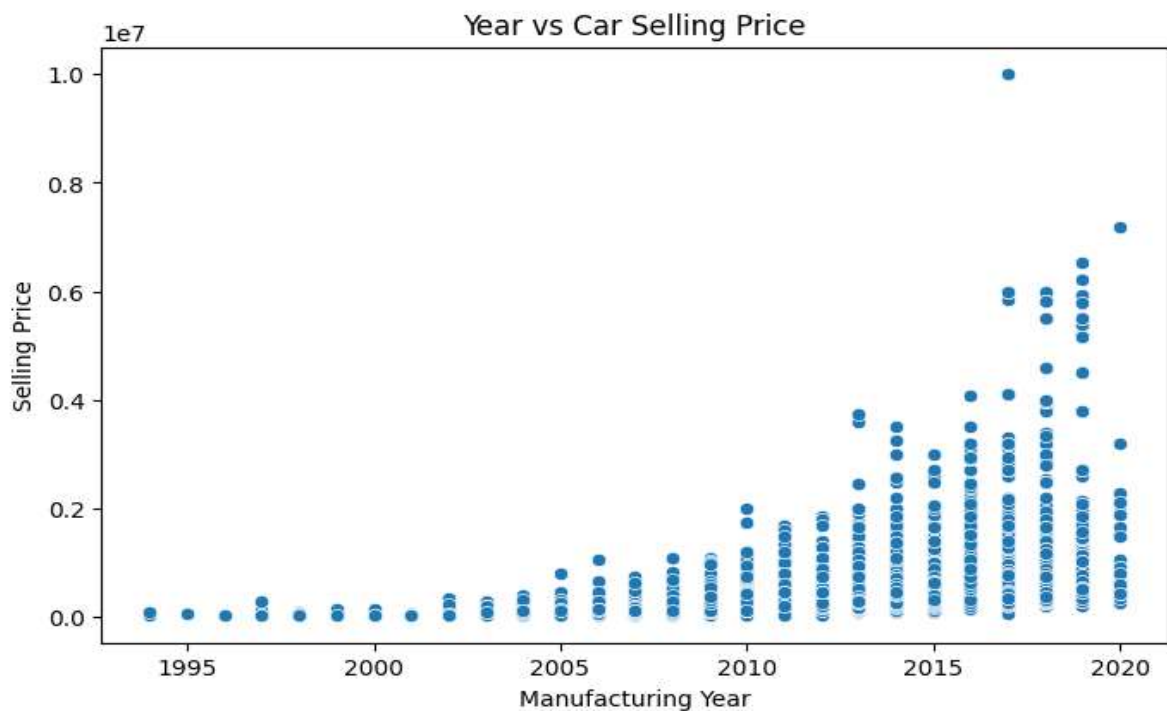


4.2 Year vs Selling Price

A scatter plot is created between manufacturing year and selling price. This helps examine whether newer or older vehicles tend to have different selling prices.

The graph uses:

- X-axis: Manufacturing Year
- Y-axis: Selling Price



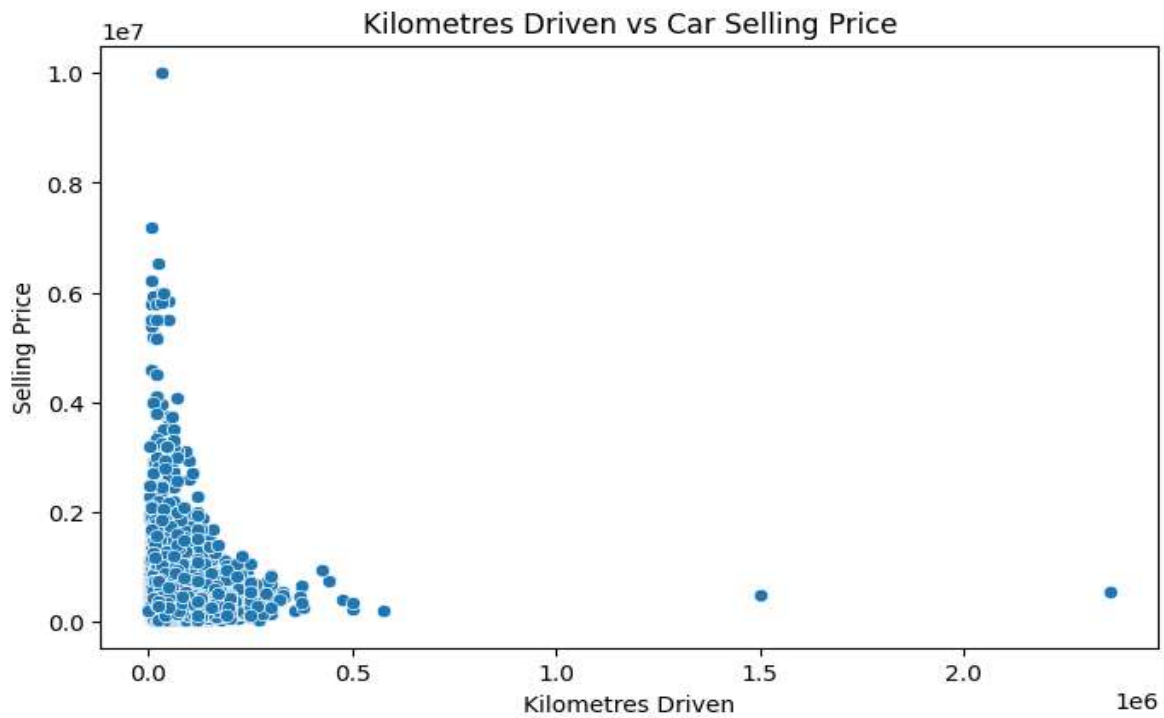
4.3 Kilometres Driven vs Selling Price

A scatter plot is used to study the relationship between kilometres driven and selling price.

Generally, this visualization helps examine whether the distance travelled by a car is associated with its selling price.

The graph uses:

- X-axis: Kilometres Driven
- Y-axis: Selling Price

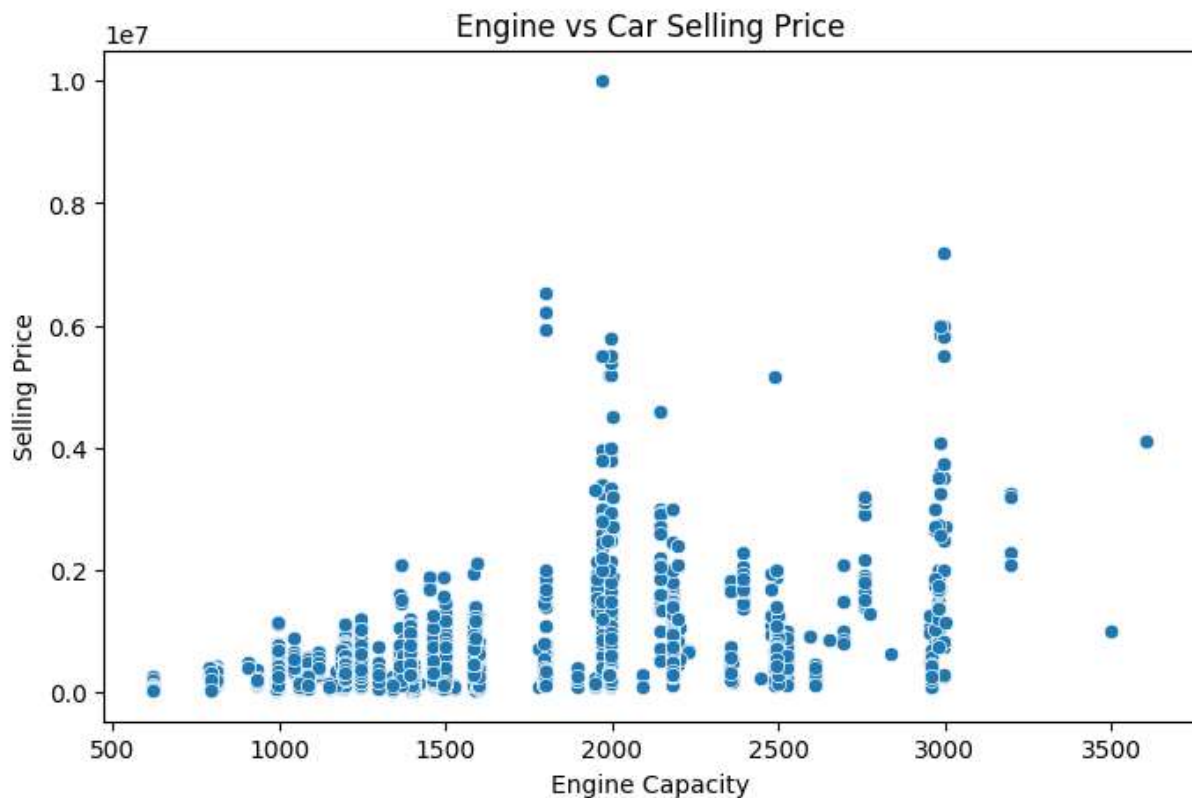


4.4 Engine vs Selling Price

A scatter plot is created between engine capacity and selling price. This helps examine whether cars with different engine capacities have different selling prices.

The graph uses:

- X-axis: Engine Capacity
- Y-axis: Selling Price



4.5 Correlation Heatmap

A correlation heatmap is created using the selected features and the target variable.

Correlation measures the strength and direction of the linear relationship between numerical variables.

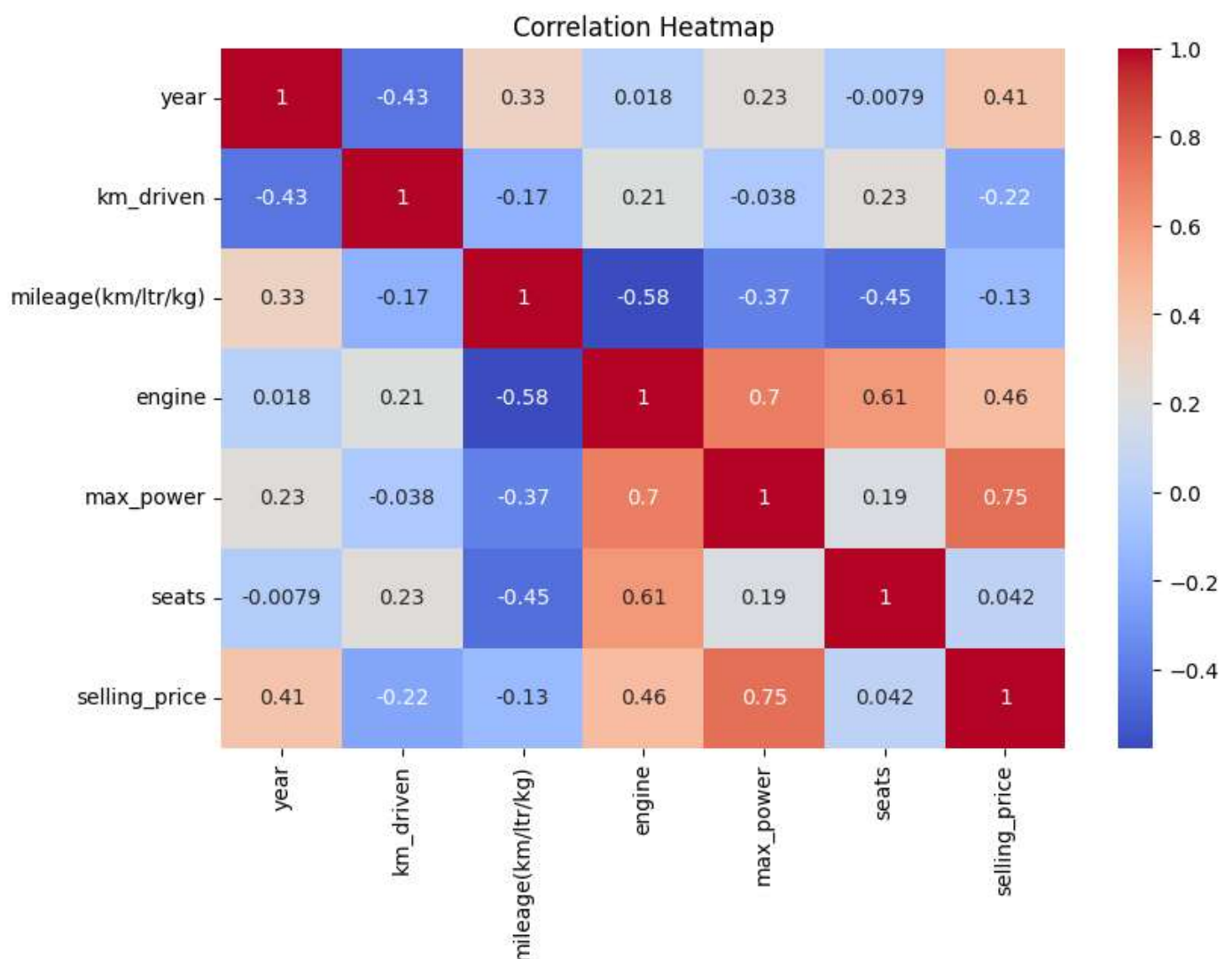
The heatmap helps identify:

- Positive relationships
- Negative relationships
- Weak relationships
- Strong relationships

The correlation matrix is calculated using:

```
X.join(y).corr()
```

The values are displayed directly on the heatmap.



5. Model Building

5.1 Machine Learning Method

The project uses **Supervised Learning**.

The selected algorithm is:

Multiple Linear Regression

Multiple Linear Regression is suitable because the project has one continuous target variable, selling price, and multiple numerical input variables.

5.2 Independent Variables

The model uses:

- Year
- Kilometres Driven
- Mileage
- Engine
- Maximum Power
- Seats

5.3 Dependent Variable

The dependent variable is:

Selling Price

5.4 Train-Test Split

The dataset is divided into training and testing data using `train_test_split()`.

The project uses:

- **70% training data**
- **30% testing data**
- `random_state = 42`

The training data is used to build the model, while the testing data is used to evaluate its performance on unseen observations.

5.5 Model Training

A Linear Regression model is created using:

`LinearRegression()`

The model is trained using:

`model.fit(X_train, y_train)`

After training, the model is used to predict prices for the test dataset.

6. Model Evaluation

Model evaluation is required to determine how well the model predicts car prices.

The project uses four evaluation measures:

1. R^2 Score
2. Adjusted R^2
3. RMSE
4. MAE

6.1 R^2 Score

R^2 , or the coefficient of determination, measures how much of the variation in the target variable is explained by the model.

A higher R^2 generally indicates that the model explains a larger proportion of the variation in the target variable.

The project calculates R^2 using:

```
r2_score(y_test, y_pred)
```

The actual value generated by the Python program should be included in the final GitHub report after running the code.

6.2 Adjusted R^2

Adjusted R^2 considers both the model performance and the number of predictors used in the model.

It is calculated using the R^2 score, number of observations and number of predictors.

6.3 RMSE

RMSE stands for **Root Mean Squared Error**.

It measures the typical size of prediction errors while giving greater weight to larger errors.

The project calculates RMSE using:

```
np.sqrt(mean_squared_error(y_test, y_pred))
```

6.4 MAE

MAE stands for **Mean Absolute Error**.

It measures the average absolute difference between the actual and predicted selling prices.

The project calculates MAE using:

mean_absolute_error(y_test, y_pred)

6.5 Model Performance Table

Evaluation Metric Result

R ² Score	0.63639285
Adjusted R ²	0.63547038
RMSE	494025.4061
MAE	286593.8691525

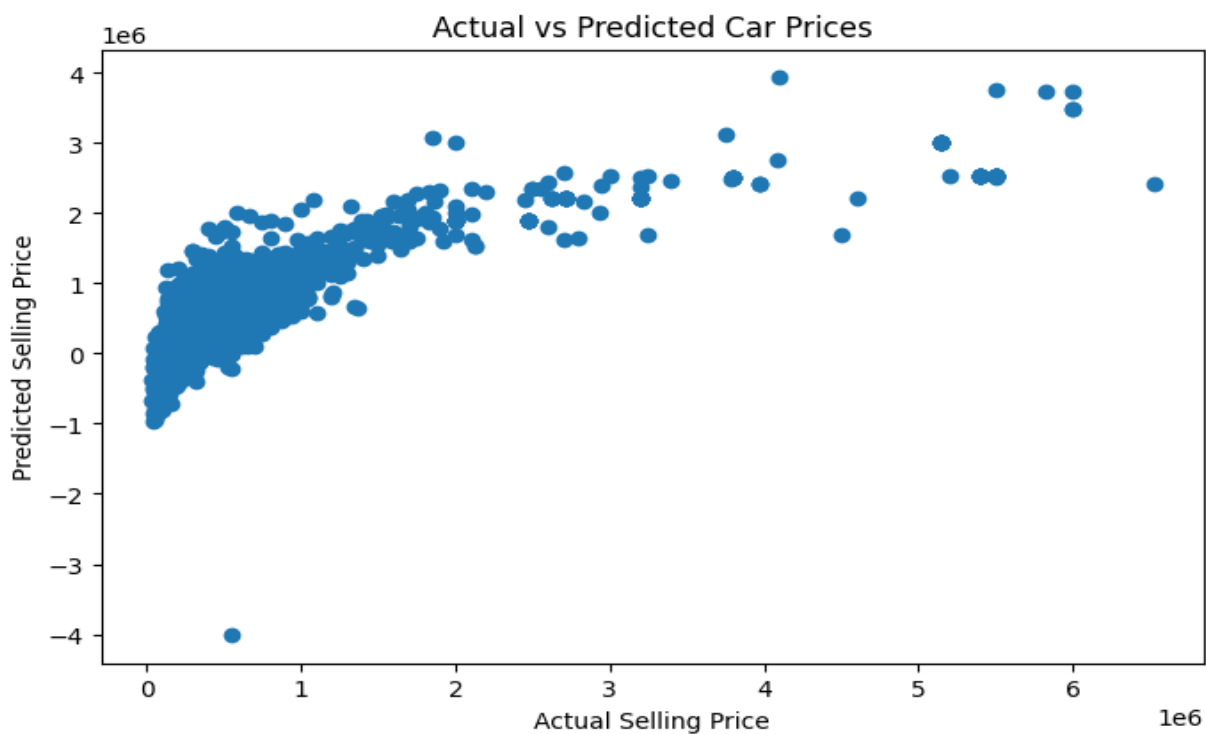
These values should be updated with the actual results obtained when the Python program is executed.

6.6 Actual vs Predicted Prices

The project creates a scatter plot comparing actual selling prices with predicted selling prices.

- X-axis: Actual Selling Price
- Y-axis: Predicted Selling Price

If predicted values are close to actual values, the points will generally be closer to a straight diagonal relationship.



7. Interpretation

The Multiple Linear Regression model uses six vehicle characteristics to estimate the selling price of a used car.

7.1 Feature Interpretation

Year

The manufacturing year represents the age of the vehicle. The regression coefficient shows how the predicted selling price changes with the year while keeping the other variables constant.

Kilometres Driven

km_driven represents the distance travelled by the car. Its coefficient indicates how the predicted price changes as kilometres driven increase, while other variables remain constant.

Mileage

Mileage represents the distance a car can travel per unit of fuel. Its coefficient indicates its estimated relationship with selling price when the other variables are kept constant.

Engine

Engine capacity represents the size/capacity of the car's engine. The coefficient indicates its relationship with predicted selling price.

Maximum Power

Maximum power represents the power output of the vehicle. The model estimates how this variable is associated with selling price while controlling for the other selected features.

Seats

The number of seats is another vehicle characteristic included in the model. Its coefficient represents its estimated relationship with selling price.

7.2 Regression Coefficients

The Python program generates a table containing the coefficient of every feature.

Feature	Coefficient
Year	38274.764991
Km Driven	-1.847537
Mileage	11185.720138

Feature	Coefficient
Engine	131.170816
Max Power	15274.971416
Seats	-68233.507457

The coefficients should be interpreted carefully because each coefficient represents the relationship with price **while the other selected variables are held constant**.

7.3 Example Prediction

The project also demonstrates prediction for a hypothetical car having:

Feature	Value
Year	2018
Kilometres Driven	50,000
Mileage	20.0
Engine	1200
Maximum Power	85
Seats	5

The trained model uses these values to generate an estimated selling price. The actual predicted value should be obtained by running the Python program.

8. Conclusion

This project demonstrates the use of **Supervised Machine Learning** for predicting the selling price of used cars.

A **Multiple Linear Regression** model was developed using six vehicle characteristics: manufacturing year, kilometres driven, mileage, engine capacity, maximum power and number of seats.

The dataset was first examined and cleaned. The max_power variable was converted into numeric form and rows with missing values in the selected variables were removed. Exploratory Data Analysis was then performed using histograms, scatter plots and a correlation heatmap.

The data was divided into training and testing sets using a 70:30 split. The Multiple Linear Regression model was trained using the training data and evaluated using **R^2 , Adjusted R^2 , RMSE and MAE**.

The project shows how statistical analysis and supervised machine learning can be applied to a practical problem such as used-car price prediction.

Limitations

The project has some limitations:

1. Only six vehicle characteristics are used for prediction.
2. The provided code does not perform a separate outlier-treatment procedure.
3. Categorical variables from the original dataset are not included in the final model.
4. Other factors such as car brand, model, fuel type, transmission and location may also influence selling price but are not used in this model.
5. Linear Regression assumes a linear relationship between the selected predictors and selling price.
6. Model performance depends on the quality and characteristics of the dataset.

Future Scope

The project can be improved in the future by:

- Including additional relevant features.
- Applying suitable categorical-variable encoding.
- Performing detailed outlier analysis.
- Comparing Multiple Linear Regression with other supervised learning algorithms.
- Applying feature selection techniques.
- Performing cross-validation.
- Creating a user-friendly application for car price prediction.